

Cloud Deployment Architecture & Partnership Document

Onekof Project Management (PM) on Ethio Cloud

DAPS PLC

August 2026

Elastic Cloud Server (ECS) Infrastructure · Single-Image Multi-Environment Strategy

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Elastic Cloud Server (ECS) Infrastructure · Single-Image Multi-Environment Strategy

Ethiopia's multi-tenant project & budget accountability platform — engineered once, deployed everywhere it is trusted to run.

DAPS PLC

Addis Ababa, Ethiopia

August 2026

Prepared for Ethio Telecom — Cloud, Networking & Security Teams

1. Executive Summary

DAPS PLC presents this document to Ethio Telecom's cloud, networking, and security teams as the complete architecture and partnership reference for deploying **Onekof PM** — Ethiopia's multi-

tenant project and budget accountability platform — on Ethio Cloud’s Elastic Cloud Server (ECS) infrastructure. It combines a company and product introduction, a technical analysis of the ECS platform, the full deployment architecture, and the specific confirmations DAPS requests so that provisioning can proceed in a single pass.

The engineering principle at the heart of this deployment is deliberate: **DAPS ships one 430 MB container image, and only one.** That identical artifact serves every market segment — government agencies on Ethio Cloud, banks in their own data centers, and NGOs and private organizations on Azure. What differs between environments is configuration, never the image. One artifact means one security audit, one patch stream, and one honest answer to “what version are you running?”

Deployment environments at a glance

Environment	Market Segment	Compliance Driver
Ethio Cloud ECS	Government agencies & ministries	Data sovereignty — data remains on Ethiopian national infrastructure

The scope of this request

DAPS owns and operates everything above the VM layer — OS hardening, container runtime, application, updates, monitoring, and backups. Zero operational burden on Ethio Telecom staff above the hypervisor, with deployment readiness within 48 hours of VM access.

2. DAPS PLC — Company Profile

DAPS PLC is an Ethiopian-registered software development and analytics company building enterprise-grade platforms for the Ethiopian market and its institutions. The company’s founding conviction is that Ethiopian organizations — public and private — deserve software engineered to international standards, designed around Ethiopian workflows, languages, payment rails, and regulatory realities, and hosted wherever their compliance obligations require.

DAPS builds and operates a portfolio of platforms spanning project and budget management, water-quality information systems, and telehealth — each developed with the same engineering discipline: security-first design, documented quality assurance, registered intellectual property, and production deployments serving real institutions.

What distinguishes DAPS

- **Ethiopian-built, internationally engineered.** DAPS’s founding team combines deep local institutional knowledge with certified cloud expertise across Microsoft Azure and AWS, applying hyperscaler-grade architecture practices to Ethiopian deployments.
- **Security and accountability as product values.** DAPS platforms are designed for environments where auditability is not a feature but a mandate — government budget oversight, financial institutions, and regulated sectors.

- **Full-lifecycle ownership.** DAPS designs, builds, deploys, and operates its platforms end-to-end, which is why it can credibly commit to zero operational burden on infrastructure partners.
- **Sovereignty-aware by design.** DAPS architectures assume from day one that different customers carry different data-residency obligations — and that the software, not the customer, should absorb that complexity.

Why Ethio Telecom

For Ethiopian public-sector customers, data sovereignty is non-negotiable: their data must reside on national infrastructure. Ethio Cloud is the natural home for that tier, and Ethio Telecom is the natural infrastructure partner. DAPS approaches this engagement not as a one-off hosting purchase but as the first workload in a longer partnership — one in which Onekof PM demonstrates, at national scale, that Ethiopian public-sector software can run sovereign, secure, and modern on Ethio Cloud.

3. Onekof PM — Product Overview

A multidisciplinary, multi-tenant platform

Onekof PM is a **multi-tenant project and budget management platform** built for Ethiopian organizations. A single deployment serves many organizations — each ministry, bank, or NGO operates in its own isolated tenant with its own users, projects, budgets, files, and audit trail, while DAPS operates one platform underneath them all. Tenants are addressed through their own subdomains (for example, a ministry reaches its workspace at its dedicated address under onekof.et), giving each institution the experience of its own system with the economics and consistency of a shared platform.

The platform is multidisciplinary by design: the same system serves the planning office tracking capital projects, the finance directorate enforcing budget discipline, the program manager coordinating field teams, and the executive requiring a real-time accountability view — across government, banking, and civil-society contexts. Its strategic framing in the public sector is explicit: Onekof PM is an accountability platform for reducing government waste and fraud, giving oversight bodies a transparent, tamper-evident record of how projects progress and how budgets are spent.

Core capabilities

Capability Area	What it delivers
Project management	Full project lifecycle — planning, task assignment, milestones, progress tracking, and cross-team coordination for institutional programs.
Budget management	Budget definition, allocation, and expenditure tracking with enforcement of spending discipline and variance visibility.
Multi-tenant organizations	Isolated tenant workspaces with per-organization users, roles, data, and subdomains — many institutions, one platform.

Role-based access control	Four-role model (OWNER, ADMIN, MEMBER, GUEST) with strict cross-tenant isolation and least-privilege defaults.
Audit & accountability	Append-only audit logs for organization, budget, and administrative actions — a tamper-evident institutional record.
Ethiopian payment rails	Native integration with Ethiopian payment processing (Telebirr, CBE via Chapa) for billing and subscriptions.
File management	Document uploads encrypted at rest with AES-256-GCM, scoped per tenant.
Notifications & email	Transactional email delivery for invitations, alerts, and workflow events.

Certification and market position

Onkof PM has achieved **INSA certification** — a significant milestone for institutional and government sales in Ethiopia, and independent validation of the platform’s security posture by the national authority. Combined with its 49-test security assessment and its sovereignty-compatible deployment model, this positions Onkof PM as an industry-leading choice for Ethiopian institutions that cannot compromise on either modern software experience or regulatory compliance.

Positioning in one sentence: Onkof PM is the accountability platform Ethiopian institutions can actually adopt: modern, multi-tenant, INSA-certified, and deployable wherever their data is required to live — national cloud, own data center, or international cloud.

4. Onkof PM Runtime Architecture

Onkof PM runs as a self-contained Docker stack: application, database, cache, and reverse proxy each in its own container, connected by an internal Docker network. Only the reverse proxy is exposed to the internet; the database and cache are never publicly reachable.

**Internet → Caddy (SSL, :443) → Onkof Web (Next.js, :3000) → PostgreSQL 15 (:5432)
 ↳ Redis 7 (:6379)**

Service	Image	Footprint	Purpose
Caddy	caddy:2-alpine	~50 MB	SSL termination, reverse proxy, security headers
Onkof Web	onekof-web:v1.0	~300 MB	Next.js application (standalone)
	postgres:15-		Primary

PostgreSQL	alpine	512 MB RAM	database (70+ tables)
Redis	redis:7-alpine	512 MB RAM	Rate limiting, caching, sessions

Two independent layers of recovery protect the platform. At the application layer, Onekof performs a daily `pg_dump` at 03:00 EAT with seven-day retention, stored on the dedicated data disk. At the infrastructure layer, DAPS uses ECS snapshots — and Cloud Server Backup where available — to capture point-in-time recovery images of the whole VM. The two layers cover different failure modes: logical data corruption on one hand, whole-VM loss on the other.

Scope note — managed Kubernetes: The ECS User Guide documents an IaaS platform: VMs, block storage, virtual networking, and load balancing. It does not document a managed Kubernetes service (the equivalent of EKS or AKS). This is not an obstacle: Onekof PM is a self-contained Docker stack operated directly by DAPS on the VM and does not depend on a managed orchestration control plane. No container-orchestration service is requested from Ethio Telecom.

5. Security and Compliance Posture

Onekof PM was built security-first, and its security properties are identical in every environment because the image is identical in every environment. This section is written so that it can be reused verbatim as the compliance summary handed to any counterparty — a ministry's security review, a bank's information-security team, or an NGO's donor-compliance assessor.

Verified security controls

- **49 security tests passed** — authentication, authorization, session management, input validation, API security, data protection, and audit logging.
- **INSA certification** — independent validation by Ethiopia's national security authority; a prerequisite-grade credential for government adoption.
- **Encryption at rest** — AES-256-GCM for all uploaded files.
- **Role-based access control** — OWNER, ADMIN, MEMBER, GUEST with strict cross-tenant isolation.
- **Password security** — scrypt hashing with timing-safe comparison.
- **Rate limiting** — on authentication, billing, and administrative endpoints.
- **Append-only audit logs** — for organization, budget, and administrative actions.
- **Automated backups** — daily `pg_dump` at 03:00 EAT, seven-day retention.

Multi-tenant isolation — the sovereignty reviewer's first question

Because government agencies are served on shared ECS infrastructure — one production VM hosting multiple ministry tenants — the first question a data-sovereignty reviewer will raise is: *"how is Ministry A's data isolated from Ministry B's?"* The answer: isolation is enforced at the application layer through the role-based access-control model, with cross-tenant isolation verified as part of the 49-

test assessment. Every data access is scoped to the requesting tenant; there is no code path by which one tenant’s users can read another tenant’s records. Sovereignty is satisfied at the infrastructure layer — all data resides on Ethio Telecom’s national infrastructure and never leaves it — and tenant isolation is satisfied at the application layer.

6. Single-Image, Multi-Environment Strategy

DAPS builds **one 430 MB Docker image** and deploys it, byte-for-byte identical, to every environment. The image is never forked, never rebuilt per customer, never specialized at build time. Everything that differs between a government ECS deployment, an on-premise bank deployment, and an Azure deployment is expressed as configuration — injected at run time — not baked into the image.

This is a deliberate rejection of the tempting alternative — maintaining separate images per environment — which would mean multiple build pipelines, multiple patch streams to track, growing configuration drift, and the ever-present risk that a fix applied to one build is forgotten in another. One image means one thing to patch, one thing to audit, and a single honest answer to “what version are you running?”

What stays identical everywhere

- The container image itself — same 430 MB artifact, same SHA digest.
- The container topology — Caddy, Onekof Web, PostgreSQL, Redis on an internal network.
- The application code, security controls, and audit-logging behavior.
- The 49-test security assessment and INSA-certified posture — one artifact, one audit, valid everywhere.

What is allowed to differ — configuration only

Config Dimension	Variation Across Environments	Mechanism
Environment variables	Per-site secrets, DB credentials, feature flags	.env file / secrets injection
DNS / hostname	onekof.et vs. bank-internal vs. Azure domain	DNS records + Caddy site block
TLS trust	Let’s Encrypt (public) vs. internal CA (air-gapped)	Caddy TLS configuration
Image delivery	Registry pull vs. SCP/USB offline load	Deployment script mode flag
Outbound integrations	Reachable (public) vs. degraded (air-gapped)	Graceful feature degradation
Backup target	Local data disk / external store	Backup configuration

Engineering disciplines this requires

1. **Graceful degradation of external integrations.** An air-gapped bank data center cannot reach Chapa, Resend, Let's Encrypt, or Sentry. The application tolerates unreachable integrations — payment and email features degrade cleanly rather than crash-looping — and TLS certificates come from an internal CA instead of a public issuer.
2. **Standardized configuration and secret delivery.** One consistent mechanism (a config template plus a per-site values file) injects secrets in every environment, so Ethio Telecom, a bank's IT team, and DAPS's Azure setup all follow the identical runbook.
3. **Version pinning as the control lever.** The only per-environment variable release-to-release is which image tag it runs. A bank on a slow change-control cycle stays on v1.0 while the government tier moves to v1.2 — no separate codebase, and a clean patch story.

7. Deployment Process on ECS

Ethio Telecom provisions resources/infrastructure and grants access to DAPS

DAPS deployment steps

1. **OS hardening.** DAPS's Ubuntu baseline: unattended security updates, host firewall aligned with the security group, SSH hardening, fail2ban, unused services disabled.
2. **Container runtime setup.** Docker installation and internal network configuration.
3. **Image delivery.** Pull the 430 MB image from ghcr.io, or load via SCP/USB if egress is restricted.
4. **Configuration injection.** Apply the government-tier .env and secrets from the standardized template.
5. **Bring up the stack and validate.** Start containers, confirm TLS issuance, verify health checks, run smoke tests.
6. **Capture a golden private image.** Snapshot the hardened, validated VM via IMS so staging, DR, or expansion environments can be rebuilt in one step.

The item most commonly missed — the EIP: A static public IPv4 is a distinct, separately-quota-governed resource on the ECS platform. If the VM is provisioned but no EIP quota is granted, the production instance cannot be reached from the public internet and go-live is blocked. DAPS asks that EIP allocation be confirmed explicitly alongside the VM approval.

Configuration detail — do not miss UDP/443: Onekof's Caddy reverse proxy serves HTTP/3 over QUIC, which uses UDP port 443 in addition to TCP 443. Because security-group rules are added port-by-port and protocol-by-protocol, it is easy to open only TCP/443 and silently break HTTP/3. Please ensure BOTH TCP/443 and UDP/443 are opened, and restrict SSH (22) to DAPS source IP ranges.

Offline fallback — no hard dependency on outbound internet: If outbound access to ghcr.io is unavailable, DAPS transfers the ~430 MB Docker image via SCP or USB. The deployment script supports both online (registry pull) and offline (image load) modes — the same capability that makes air-

gapped bank deployments possible from the identical image. The government ECS deployment can proceed even in a restricted-egress configuration.

8. Workflow and Dependencies

Zero operational burden above the VM layer: Ethio Telecom's responsibility ends at delivering correctly-networked VMs with adequate quota. DAPS handles everything above the hypervisor and commits to deployment readiness within 48 hours of receiving VM access.

Milestone	Target	Dependency
VM provisioning	Upon Ethio Telecom's approval	Ethio Telecom approval + quota
OS hardening + Docker setup	Within 5 days of access	VM access
Staging deployment + testing	5 days after access	Network connectivity
Production go-live	Following successful staging	.et domain (or IP fallback)

9. Next Steps and Partnership with Ethio Telecom

Confirmations requested from Ethio Telecom

1. **Confirm resource quota.** Verify the aggregate quota — vCPU, RAM, EVS capacity, ECS instances, and the EIP specifically.
2. **Confirm security-group rules.** Both TCP/443 and UDP/443 opened; SSH (22) restricted to DAPS source IPs.
3. **Confirm outbound policy.** Whether outbound is unrestricted or locked down; if locked down, allow-list the required destinations (ghcr.io, api.chapa.co, api.resend.com, acme-v02.api.letsencrypt.org, archive.ubuntu.com).
4. **Confirm staging egress path.** Public (own EIP) or private-behind-NAT, so image delivery mode can be set correctly.
5. **Confirm managed-Kubernetes availability.** Not required now, but useful for future scaling; does Ethio Cloud offer a managed container-orchestration service beyond raw ECS VMs?
6. **Assist with .et domain registration.** Registration of onekof.et through Ethio Telecom where possible; IP-based fallback available.

The broader partnership — after go-live

The government ECS deployment is the first phase of a longer relationship. With INSA certification achieved and the platform positioned as an accountability tool for reducing government waste

and fraud, DAPS sees these natural next steps in partnership with Ethio Telecom:

- **Federal ministry beachhead expansion.** Onboarding additional ministries as tenants on the shared platform, demonstrating the multi-tenant model at national scale.
- **A repeatable government-deployment blueprint.** Golden private image plus standardized runbook, so each new agency environment is provisioned predictably and auditably.
- **Ethio Cloud as the sovereign platform of record.** Positioning Ethio Cloud as the hosting standard for Ethiopian public-sector software, with Onkof PM as an anchor workload demonstrating the platform's readiness.
- **Capacity planning for scale.** As tenant count grows: horizontal scaling with additional VMs behind Elastic Load Balance, and higher quota tiers.

What DAPS commits to Ethio Telecom

- A single, auditable, INSA-certified image — one artifact, one audit, deployed identically.
- Zero operational burden above the VM layer, with 48-hour deployment readiness.
- Transparent version management — always clear what is running where.
- A reusable compliance summary (Section 5) for any government security review.
- A partnership posture that treats Ethio Cloud as the sovereign platform of record for Ethiopian public-sector deployments.

Appendix A: Ethio Telecom Cloud Infrastructure — Migration Planning

Context

Onkof PM received **INSA certification**. To serve Ethiopian government institutions, we must deploy on sovereign infrastructure — Ethio Telecom's cloud (Elastic Cloud Server / ECS).

Our Docker image (ghcr.io/olitamrat/onekof-web:1.0.0) was built and ready. The deployment pipeline (deploy-et.sh, CI/CD, Caddyfile, backup scripts) is complete and tested.

Purpose of this section: Summarize what Ethio Telecom offers, what we need, the gap analysis, and the selected deployment option.

What Onkof PM Requires

Onkof runs as a Docker stack with four containers:

Container	Role	Resource Profile
onekof-web	Next.js 14 standalone (Node.js)	CPU-moderate, 1-2 GB RAM
	PostgreSQL 15 (with	RAM-heavy (query

postgres	pgvector)	cache, connections), disk I/O
redis	Session cache, rate limiting	Light (~128 MB)
caddy	Reverse proxy, auto- TLS	Light (~64 MB)

Resource requirements

Resource	Minimum	Comfortable	Arch Doc (Original)
vCPU	2	4	8
RAM	8 GB	16–32 GB	32 GB
OS Disk	30 GB	30 GB	100 GB
Data Disk	25 GB	50 GB	500 GB
Bandwidth	2 Mbps	4 Mbps	Not specified
Public IP	Required	Required	Required

Why RAM Optimized over CPU Optimized? Onekof is a database-backed web application, not a compute-intensive workload. PostgreSQL performance scales primarily with RAM (buffer cache, sort memory, connection pooling). Next.js server-side rendering is moderate CPU. RAM-optimized plans are the correct choice.

Storage estimate at launch

Usage	Estimated Size
Docker images (4 containers)	~2 GB
Daily backup archive (7 days)	~5–10 GB
OS and logs	~5 GB
Total at launch	~20–35 GB (fits in 50 GB)

Critical storage gap: Ethio Telecom’s standard plans include a maximum of 50 GB SSD data storage. The original architecture document specified 500 GB. At launch, estimated usage fits within 50 GB, but EVS add-on volumes should be confirmed with ETZCloudSupport@ethiotelecom.et for future scaling.

Selected Deployment Option

Scenario 2: Lean + Managed Backup — RECOMMENDED

126,815 ETB / 6 months (~\$169/month)

Component	Configuration	Cost (ETB / 6 months)
Production VM	CSRAMOPT05 (4 vCPU, 32 GB RAM, 50 GB SSD, 4 Mbps)	107,047
Managed Backup	ETZ Daily Backup (100 GB, 7-day retention)	19,768
Staging	Vercel (free, existing — synthetic data only)	0

Firewall	ufw + Caddy (built-in, INSA-hardened)	0
Total		126,815

Dual-layer backup strategy

Layer 1: Ethio Telecom’s managed Daily Backup service performs automated full VM snapshots with 7-day retention — managed and guaranteed by Ethiopia’s national telecom provider.

Layer 2: Our application runs independent daily PostgreSQL backups with 7-day rolling retention for granular database recovery.

In a disaster, Ethio Telecom restores the full environment in minutes while we can independently verify data integrity at the application level.

Alternative scenarios for reference

Scenario	Configuration	Cost (ETB / 6 mo)	~USD/month
1: Lean Launch	CSRAMOPT05 + backup-db.sh only, staging on Vercel	107,047	~\$143
2: Lean + Managed Backup	CSRAMOPT05 + ETZ Daily Backup, staging on Vercel	126,815	~\$169
3: Full 2-VM + Backup	CSRAMOPT06 (prod) + CSRAMOPT01 (staging) + ETZ Daily	196,185	~\$262
4: Maximum + All Add-Ons	CSRAMOPT08 (prod) + CSRAMOPT02 (staging) + ETZ Daily	368,674	~\$492

For context: A Monday.com license for 100 government users costs ~\$9,600/year. Our entire infrastructure — including managed backup by Ethio Telecom — costs less than 27% of that. The \$26/month backup add-on is the difference between a startup pitch and a government-ready vendor.

Staging on Vercel — sovereignty assessment

Data sovereignty applies to production data — real government users, projects, documents. Staging contains only synthetic test data created by DAPS developers. Production (onekof.et) and staging (onekof.com) are completely separate databases with zero data flow between them. No real government data touches Vercel. This is industry standard — AWS GovCloud and Azure Government both use non-sovereign staging environments.

Appendix B: ETZ-Backup Catalog and Add-On Services

ETZ-Backup (Full Catalog)

Scenario 2 includes the **Daily** tier (highlighted). Other tiers shown for reference:

Tier	Frequency	Retention	ETB / 6 Months
Weekly	Every weekend	4 copies (4 weeks)	11,695
Daily (Selected)	Every day	7 copies (7 days)	19,768
Hourly	Every hour	12 copies (12 hours)	33,224

Selected RAM-Optimized Server Plans (CSRAMOPT)

Source: myportal.ethiotelecom.et — “RAM & CPU Optimized Cloud Server” category. All plans are semi-annual (6-month) billing. All include: 30 GB SSD OS storage, dedicated public IP, weekly backup (1 copy), SSH access, 24/7 basic support via ETZCloudSupport@ethiotelecom.et.

SKU	vCPU	RAM	Data SSD	Bandwidth	ETB / 6 Months	~USD / Month
CSRAMOPT07	8	64 GB	50 GB	2 Mbps	242,641	~\$324
CSRAMOPT05	4	32 GB	50 GB	4 Mbps	107,047	~\$143
CSRAMOPT03	2	16 GB	25 GB	8 Mbps	77,302	~\$103
CSRAMOPT01	2	16 GB	25 GB	2 Mbps	60,202	~\$80

Selected CPU-Optimized Server Plans (CSCPUOPT)

SKU	vCPU	RAM	Data SSD	Bandwidth	ETB / 6 Months	~USD / Month
CSCPUOPT12	8	16 GB	50 GB	8 Mbps	116,215	~\$155
CSCPUOPT07	4	8 GB	50 GB	2 Mbps	58,013	~\$77
CSCPUOPT01	1	2 GB	30 GB	2 Mbps	19,206	~\$26

Action Items

- Approve Scenario 2** — CSRAMOPT05 (107,047 ETB) + ETZ Daily Backup (19,768 ETB) = 126,815 ETB / 6 months.
- Confirm staging approach** — Staging uses only synthetic data on Vercel, no sovereignty impact on production.

3. **Cloud Firewall** — Skip at launch (use ufw + Caddy + INSA middleware). Revisit only if contractually required by a specific government client.
4. **Storage expansion** — Contact ETZCloudSupport@ethiotelecom.et about EVS volume pricing for future scaling beyond 50 GB.
5. **Timing** — Order now (August 2026) to maximize the INSA certification window, or wait for a specific trigger (e.g., signed LOI).

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